

Economics 326
Methods of Empirical Research in Economics
Lecture 14: Hypothesis testing in the multiple
regression model, Part 2

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May 5, 2010

Multiple restrictions

- Consider the model:

$$\ln(Wage_i) = \beta_0 + \beta_1 Experience_i + \beta_2 Experience_i^2 + \beta_3 PrevExperience_i + \beta_4 PrevExperience_i^2 + \beta_5 Education_i + U_i,$$

where *Experience* is the experience at current job, and *PrevExperience* is the previous experience.

- Suppose that we want to test the null hypothesis that, after controlling for the experience at current job and education, the previous experience has no effect on wage:

$$H_0 : \beta_3 = 0, \beta_4 = 0.$$

- We have two restrictions on the model parameters.
- The alternative hypothesis is that at least one of the coefficients, β_3 or β_4 , is different from zero:

$$H_1 : \beta_3 \neq 0 \text{ or } \beta_4 \neq 0.$$

t -statistics and multiple restrictions

- ▶ Let T_3 and T_4 be the t -statistics associated with the coefficients of $PrevExperience$ and $PrevExperience^2$:

$$T_3 = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)} \text{ and } T_4 = \frac{\hat{\beta}_4}{SE(\hat{\beta}_4)}.$$

- ▶ We can use T_3 and T_4 to test significance of β_3 and β_4 separately by using two separate size α tests:
 - ▶ Reject $H_{0,3} : \beta_3 = 0$ in favor of $H_{1,3} : \beta_3 \neq 0$ when $|T_3| > t_{n-k-1, 1-\alpha/2}$.
 - ▶ Reject $H_{0,4} : \beta_4 = 0$ in favor of $H_{1,4} : \beta_4 \neq 0$ when $|T_4| > t_{n-k-1, 1-\alpha/2}$.

t -statistics and multiple restrictions

- ▶ Rejecting $H_0 : \beta_3 = 0, \beta_4 = 0$ in favor of $H_1 : \beta_3 \neq 0$ or $\beta_4 \neq 0$ when at least one of the two coefficients is significant at level α , i.e. when

$$|T_3| > t_{n-k-1, 1-\alpha/2} \text{ or } |T_4| > t_{n-k-1, 1-\alpha/2},$$

is not a size α test!

- ▶ Recall that if A and B are two sets then $(A \cap B) \subset A$ and therefore $P(A \cap B) \leq P(A)$.
- ▶ When $\beta_3 = \beta_4 = 0$:

$$\begin{aligned} & P(\text{Reject } H_{0,3} \text{ or } H_{0,4}) = \\ &= P(|T_3| > t_{n-k-1, 1-\alpha/2} \text{ or } |T_4| > t_{n-k-1, 1-\alpha/2}) \\ &= P(|T_3| > t_{n-k-1, 1-\alpha/2}) + P(|T_4| > t_{n-k-1, 1-\alpha/2}) \\ &\quad - P(|T_3| > t_{n-k-1, 1-\alpha/2} \text{ and } |T_4| > t_{n-k-1, 1-\alpha/2}) \\ &= \alpha + \alpha - P(|T_3| > t_{n-k-1, 1-\alpha/2} \text{ and } |T_4| > t_{n-k-1, 1-\alpha/2}) \\ &\geq \alpha. \end{aligned}$$

Testing multiple exclusion restrictions

- Consider the model

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_q X_{q,i} + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

Suppose that we want to test that the first q regressors have no effect on Y (after controlling for other regressors).

- The null hypothesis has q exclusion restrictions:

$$H_0 : \beta_1 = 0, \beta_2 = 0, \dots, \beta_q = 0.$$

- The alternative hypothesis is that at least one of the restrictions in H_0 is false:

$$H_1 : \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \text{ or } \dots \text{ or } \beta_q \neq 0.$$

F-statistic

- ▶ The idea of the test is to compare the **fit** of the unrestricted model with that of the null-restricted model.
- ▶ Let SSR_{ur} denote the Residual Sum-of-Squares of the unrestricted model

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_q X_{q,i} + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

- ▶ The restricted model given $H_0 : \beta_1 = 0, \dots, \beta_q = 0$ is

$$Y_i = \beta_0 + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

- ▶ Let SSR_r denote the Residual Sum-of-Squares of the restricted model .
- ▶ Consider the following statistic:

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)}.$$

- ▶ Note that $q =$ **number of restrictions**;
- ▶ $n - k - 1 =$ **unrestricted residual df**, where k is the number of regressors in the **unrestricted model**.

F-statistic

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)}.$$

- ▶ Since SSR can only increase when you drop some regressors,

$$SSR_r - SSR_{ur} \geq 0,$$

and therefore $F \geq 0$.

- ▶ If the null restrictions are true, the excluded variables do not contribute to explaining Y (in population), and therefore we should expect that $SSR_r - SSR_{ur}$ is small and F is close to zero.
- ▶ If the null restriction are false, the imposed restriction should substantially worsen the fit, and we should expect that $SSR_r - SSR_{ur}$ is large and F is far from zero.
- ▶ Thus, we should reject H_0 when $F > c$ where c is some positive constant.

F test

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)}.$$

- ▶ We should reject H_0 when $F > c$.
- ▶ There is a probability that $F > c$ even when H_0 is true, thus we need to choose c so that $P(F > c | H_0 \text{ is true}) = \alpha$.
- ▶ It turns out that when H_0 is true, the F -statistic has F distribution with two parameters: the numerator df (q) and the denominator df ($n - k - 1$):

$$F \sim F_{q, n-k-1}.$$

- ▶ Similarly to the standard normal and t distributions, the F distribution has been tabulated and its critical values are available in statistical tables and statistical software such as Stata.

F test

When H_0 is true,

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)} \sim F_{q, n-k-1}.$$

- ▶ Let $F_{q, n-k-1, \tau}$ be the τ -quantile of the $F_{q, n-k-1}$ distribution.
- ▶ A size α test $H_0 : \beta_1 = 0, \dots, \beta_q = 0$ against $H_1 : \beta_1 \neq 0$ or $\dots$ or $\beta_q \neq 0$ is

Reject H_0 when $F > F_{q, n-k-1, 1-\alpha}$.

- ▶ One can find the p -value by finding τ such that $F = F_{q, n-k-1, 1-\tau}$. The p -value is equal to τ .

F distribution in Stata

- ▶ To compute F critical values use

`disp invFtail( $q$ ,  $n - k - 1$ ,  $\alpha$ ).`

- ▶ To compute p -values from F distribution use

`disp Ftail( $q$ ,  $n - k - 1$ ,  $F$ ).`

Example

- Consider the model:

$$\ln(Wage_i) = \beta_0 + \beta_1 Experience_i + \beta_2 Experience_i^2 + \beta_3 PrevExperience_i + \beta_4 PrevExperience_i^2 + \beta_5 Education_i + U_i.$$

- We test

$$H_0 : \beta_3 = 0, \beta_4 = 0 \text{ against } H_1 : \beta_3 \neq 0 \text{ or } \beta_4 \neq 0.$$

- $q = 2$.
- $\alpha = 0.05$.

Example: the unrestricted model

```
. regress lnWage Experience Experience2 PrevExperience PrevExperience2 Education
```

Source	SS	df	MS	Number of obs =	526
Model	51.3318741	5	10.2663748	F(5, 520) =	55.04
Residual	96.9978773	520	.186534379	Prob > F =	0.0000
Total	148.329751	525	.28253286	R-squared =	0.3461
				Adj R-squared =	0.3398
				Root MSE =	.4319

lnWage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Experience	.0471914	.0068074	6.93	0.000	.0338179	.0605649
Experience2	-.0008518	.0002472	-3.45	0.001	-.0013374	-.0003662
PrevExperi~e	.0168997	.0047331	3.57	0.000	.0076013	.0261981
PrevExperi~2	-.0003727	.0001208	-3.09	0.002	-.00061	-.0001354
Education	.0887704	.0072131	12.31	0.000	.0745999	.1029408
_cons	.2368427	.10287	2.30	0.022	.0347509	.4389346

- ▶ $SSR_{ur} = 96.9978773$.
- ▶ $n - k - 1 = 526 - 5 - 1 = 520$.

Example: the restricted model

```
. regress lnWage Experience Experience2 Education
```

Source	SS	df	MS	Number of obs	=	526
Model	48.8668114	3	16.2889371	F(3, 522)	=	85.49
Residual	99.46294	522	.190542031	Prob > F	=	0.0000
Total	148.329751	525	.28253286	R-squared	=	0.3294
				Adj R-squared	=	0.3256
				Root MSE	=	.43651

lnWage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Experience	.0510784	.0067937	7.52	0.000	.037732	.0644248
Experience2	-.0009941	.0002463	-4.04	0.000	-.001478	-.0005103
Education	.0852822	.0068978	12.36	0.000	.0717313	.0988331
_cons	.3688491	.0908138	4.06	0.000	.1904437	.5472544

► $SSR_r = 99.46294$.

Example: F statistic and test

- To compute the statistic:

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)} = \frac{(99.46294 - 96.9978773) / 2}{96.9978773 / (526 - 5 - 1)} \approx 6.61.$$

- The critical value:

```
. disp invFtail(2,520,0.05)  
3.0130572
```

- The test: $6.61 > 3.0130572$ and at 5% significance level we reject H_0 that previous experience has no effect on wage.
- The p -value:

```
. disp Ftail(2,520,6.61)  
.00146284
```

$\Rightarrow$ We reject H_0 for any $\alpha > 0.00146284$.

Example: Stata test command

- ▶ Instead of running two models, restricted and unrestricted, one can use the Stata test command after estimation of the unrestricted model.
- ▶ To test that previous experience has no effect:

```
. test (PrevExperience=0) (PrevExperience2=0)
```

- ▶ The output of this command is:

```
( 1) PrevExperience = 0  
( 2) PrevExperience2 = 0  
F( 2, 520) = 6.61  
Prob > F = 0.0015
```

Example: Stata test command

- ▶ To test that the coefficient on previous experience equal to the coefficient on experience and the coefficient on previous experience squared is zero:

```
. test (Experience==PrevExperience2) (PrevExperience2=0)
```

- ▶ The output is:

```
( 1) Experience - PrevExperience2 = 0
```

```
( 2) PrevExperience2 = 0
```

```
      F( 2, 520) = 31.94
```

```
      Prob > F = 0.0000
```


F and R^2

- ▶ Let R_{ur}^2 denote the R^2 corresponding to the unrestricted model:

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_q X_{q,i} + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

- ▶ Let R_r^2 denote the R^2 corresponding to the restricted model:

$$Y_i = \beta_0 + \beta_{q+1} X_{q+1,i} + \dots + \beta_k X_{k,i} + U_i.$$

- ▶ The two models have the same dependent variable and therefore the same Total Sum-of-Squares:

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2 = SST_{ur} = SST_r.$$

- In this case, we can write then

$$\begin{aligned} F &= \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)} \\ &= \frac{\left(\frac{SSR_r}{SST} - \frac{SSR_{ur}}{SST} \right) / q}{\frac{SSR_{ur}}{SST} / (n - k - 1)} \\ &= \frac{(1 - R_r^2 - (1 - R_{ur}^2)) / q}{(1 - R_{ur}^2) / (n - k - 1)} \\ &= \frac{(R_{ur}^2 - R_r^2) / q}{(1 - R_{ur}^2) / (n - k - 1)}. \end{aligned}$$

F test: more examples

- Suppose that you want to test $H_0 : \beta_1 = 1$ against $H_1 : \beta_1 \neq 1$ in

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i.$$

- The restricted model is

$$Y_i = \beta_0 + X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i,$$

or

$$Y_i - X_{1,i} = \beta_0 + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i.$$

1. Generate a new dependent variable $Y_i^* = Y_i - X_{1,i}$.
2. Regress Y^* against a constant, $X_2, \dots, X_k$ to obtain SSR_r .
3. Estimate the unrestricted model to obtain SSR_{ur} .
4. Compute $F = \frac{(SSR_r - SSR_{ur})/1}{SSR_{ur}/(n-k-1)}$.

F test: more examples

- Suppose that you want to test $H_0 : \beta_1 + \beta_2 = 1$ against $H_1 : \beta_1 + \beta_2 \neq 1$ in

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i.$$

- The restricted model is

$$Y_i = \beta_0 + (1 - \beta_2) X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i} + U_i,$$

or

$$Y_i - X_{1,i} = \beta_0 + \beta_2 (X_{2,i} - X_{1,i}) + \dots + \beta_k X_{k,i} + U_i.$$

1. Generate a new dependent variable $Y_i^* = Y_i - X_{1,i}$.
2. Generate a new regressor $X_2^* = X_{2,i} - X_{1,i}$.
3. Regress Y^* against a constant, $X_2^*, X_3, \dots, X_k$ to obtain SSR_r .
4. Estimate the unrestricted model to obtain SSR_{ur} .
5. Compute $F = \frac{(SSR_r - SSR_{ur})/1}{SSR_{ur}/(n-k-1)}$.

Relationship between F and t statistics

- ▶ The F statistic can also be used for testing a single restriction.
- ▶ In the case of a single restriction, the F test and t test lead to the same outcome because

$$t_{n-k-1}^2 = F_{1,n-k-1}.$$

Test of model significance

- Consider the model

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_k X_{k,i} + U_i.$$

- Suppose that you want to test that none of the regressors explain Y :

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0 \text{ (} k \text{ restrictions) against}$$

$$H_1 : \beta_j \neq 0 \text{ for some } j = 1, \dots, k.$$

- The restricted model is given by

$$Y_i = \beta_0 + U_i,$$

and since $\hat{\beta}_0 = \bar{Y}$ in this model,

$$SSR_r = \sum_{i=1}^n (Y_i - \bar{Y})^2 = SST \text{ and } SSR_{ur} = SSR.$$

Test of model significance

- The F statistic for model significance test is

$$\begin{aligned} F &= \frac{(SSR_r - SSR_{ur}) / k}{SSR_{ur} / (n - k - 1)} \\ &= \frac{(SST - SSR) / k}{SSR / (n - k - 1)} \\ &= \frac{SSE / k}{SSR / (n - k - 1)} \\ &= \frac{R^2 / k}{(1 - R^2) / (n - k - 1)}. \end{aligned}$$

Test of model significance

- The F statistic for the model significance test and its p -value is reported by Stata as in the top part of the regression output.

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